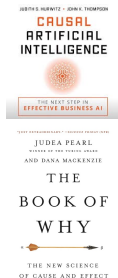


Lesson 10.1: Causal Discovery

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References:

- Pearl: *“Causality: Models, Reasoning, and Inference”* (2009)
- Peters et al.: *“Elements of Causal Inference”* (2017)
- Spirtes et al.: *“Causation, Prediction, and Search”* (2000)
- Glymour et al.: *“Review of Causal Discovery Methods Based on Graphical Models”* (2020)



- *The Discovery Problem: When and Why It Works*
 - Inferring Causal Structure from Observational Data
 - Identifiability and Causal Sufficiency
 - Practical Limitations of Automated Discovery
- When to Use Discovery vs. Domain Knowledge
- Discovery Algorithm Families
- Challenges and Validation

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Causal Discovery: Motivation

- So far we have **assumed** a causal DAG was given (e.g., drawn by a domain expert)
- **Question**
 - What if we *do not* know the DAG?
 - Can we learn causal structure directly from data?
- **Causal discovery** (a.k.a. *structure learning*)
 - Recover the causal DAG from observational and optionally interventional data
- **Why it matters**
 - Experiments are expensive, slow, or unethical
 - E.g., cannot randomize exposing humans to asbestos
 - Observational data is plentiful but *ambiguous*
 - Goal: extract causal hypotheses that can be tested later

Causal Discovery: Definition

- **Causal discovery** is the process of inferring from a dataset sampled from an unknown data-generating process:

$$\mathcal{D} \sim \Pr(X_1, \dots, X_n)$$

a graph G (or equivalence class) describing causal relations among $X_1, \dots, X_n$ such that:

$$\Pr(X_1, \dots, X_n) = \prod_{i=1}^n \Pr(X_i \mid Pa_G(X_i))$$

where $Pa_G(X_i)$ are the parents of X_i in G

- **Example:**
 - Macroeconomic drivers
 - From time-series data, infer causal links between inflation, unemployment, and interest rates
 - Education $\rightarrow$ income
 - From observational survey data, estimate whether education causally affects earnings (vs confounding)
 - Peer influence in social networks
 - From user behavior data, infer whether friends influence each

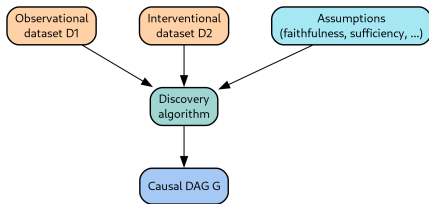
Observational vs. Interventional Data

	Observational Data	Interventional Data
Definition	Data collected by <i>watching</i> the system	Data collected under <i>do</i> -interventions
Manipulation	No variable is manipulated	Variables are actively manipulated
Example	Hospital records of patients	Randomized Controlled Trial (RCT)
Pros	Abundant and cheap No ethical constraints	Identifies causal direction Breaks spurious correlations
Cons	Cannot distinguish cause vs correlation Hidden confounders	Expensive, slow, often infeasible Ethical issues (e.g., smoking trials)

- Discovery algorithms combine both when possible

The Discovery Problem in One Picture

- We observe a table of measurements
- We do not know the DAG
- **Goal**: recover arrows among variables



- **Key insight**
 - Discovery is always based on *assumptions*
 - No algorithm recovers structure from data alone without some prior axiom

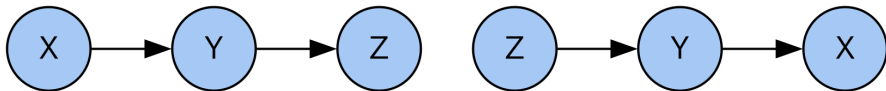
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Key Assumptions for Causal Discovery

- Causal discovery rests on three main assumptions:
 1. **Causal Markov condition**
 - Each variable is independent of its non-descendants given its parents
 - Equivalent to the factorization $\Pr(X) = \prod_i \Pr(X_i \mid Pa(X_i))$
 2. **Faithfulness**
 - All conditional independencies in the data come from the DAG structure, not from *numerical coincidences* (no cancelling paths)
 3. **Causal sufficiency**
 - No hidden common causes (unobserved confounders)
 - All relevant variables are measured
- Violating any of these can lead to an **incorrect** recovered DAG

Markov Equivalence Classes

- Can observational data distinguish every DAG?
 - **No**
 - Different DAGs may encode the same conditional independencies
- **Markov equivalent DAGs** share the same:
 - Skeleton (undirected edges)
 - v-structures (colliders $X \rightarrow Z \leftarrow Y$)
- The best an observational-only algorithm can return is a **CPDAG** (Completed Partially Directed Acyclic Graph) that represents the equivalence class



- Both imply $X \perp Z \mid Y$; indistinguishable from observational data

Identifiability: When Can We Recover the DAG?

- **Identifiable from observational data alone**
 - Only the *Markov equivalence class* (a CPDAG)
 - Directions of some edges remain ambiguous
- **Fully identifiable under stronger assumptions**
 - *Linear non-Gaussian noise* (LiNGAM): direction recovered uniquely
 - *Additive noise models* with nonlinear functions
 - *Post-nonlinear* models (Zhang et al., 2009)
- **Identifiable with interventional data**
 - Each intervention cuts incoming edges to the target variable
 - Enough interventions recover the full DAG
- **Rule of thumb**: more structure (non-Gaussianity, nonlinearity, interventions) $\Rightarrow$ more identifiability

Faithfulness: A Subtle Assumption

- **Faithfulness**: the observed distribution has *no extra* independencies beyond those implied by the DAG
- **Example of violation**
 - Path cancellation
 - Suppose $X \rightarrow Y$ and $X \rightarrow Z \rightarrow Y$
 - If the two paths' effects are equal in magnitude and opposite in sign, $X \perp Y$ in the data even though X causes Y
 - Algorithm would conclude *no edge*
- In practice, exact cancellations are *measure zero*, but near-cancellations in finite samples cause errors
- Discovery is:
 - Statistically correct in the limit
 - Fragile with small data

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Practical Limitations

- **Sample complexity**
 - Conditional independence tests need many samples
 - Error rate scales with number of tests: $O(n^2)$ pairs of variables
- **Hidden confounders**
 - Break causal sufficiency
 - Produce spurious edges or wrong directions
- **Selection bias**
 - Sampling mechanism depends on variables of interest
 - E.g., only patients sick enough to visit hospital are measured
- **Finite-sample faithfulness**
 - Weak dependencies look like independencies
 - Statistical power drops rapidly with conditioning set size

Practical Limitations (cont.)

- **Temporal information is usually ignored**
 - Yet time is the strongest causal prior
 - Some algorithms (Granger-causality, DYNOTEARS) exploit it explicitly
- **Non-stationarity and heterogeneity**
 - Distribution shifts across subpopulations break i.i.d. assumptions
- **Computational complexity**
 - Search space grows *super-exponentially* in the number of variables n :

$$|\{\text{DAGs over } n \text{ nodes}\}| \approx 2^{O(n^2)}$$

- PC algorithm: $O(n^q)$ where q is max degree
 - NP-hard in general for score-based search
- **Bottom line:** discovery is a *hypothesis generator*, not a truth oracle

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Discovery vs. Domain Knowledge

- When should we *run an algorithm* vs *ask an expert*?

Situation	Recommended approach
Well-studied domain (physics, biology)	Expert-drawn DAG, verified by data
Exploratory science (genomics)	Discovery algorithm as hypothesis generator
Policy / medicine (high stakes)	Expert DAG + RCTs for key edges
Large systems (hundreds of variables)	Discovery to prune, expert to refine
Clear temporal order	Time-based + discovery

- **Key takeaway:** discovery and expertise are *complements*, not substitutes

Discovery as Hypothesis Generation

- **Discovery algorithms output *candidate DAGs*, not truth**
 - Output depends on assumptions that may not hold
 - Multiple DAGs in the equivalence class are equally compatible
- Use discovery to:
 - Generate **plausible structures** to examine
 - Highlight **surprising edges** worth investigating
 - Flag **missing confounders** (variables with many high-correlation edges)
- Do *not* use discovery to:
 - Make final policy decisions without expert review
 - Replace RCTs for treatment-effect estimation
 - Claim causation from purely observational pipelines
- **Example:** Sachs et al. (2005) reconstructed a known protein-signaling network from single-cell data → validated discovery as a *scientific* tool

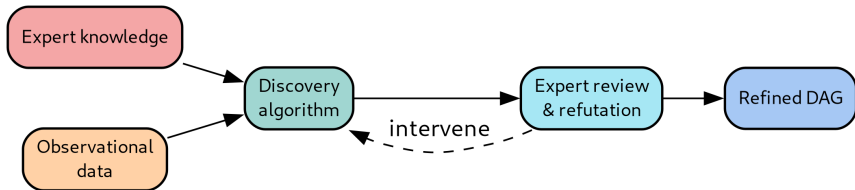
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Using Domain Knowledge as Constraints

- Partial domain knowledge can **drastically** reduce the search space
- Common forms of prior knowledge:
 - **Forbidden edges**: “variable X cannot cause variable Y ”
 - **Required edges**: “we know $X \rightarrow Y$ from physics”
 - **Tiered ordering**: variables in tier k cannot cause tier $< k$
 - E.g., *demographics* $\rightarrow$ *behavior* $\rightarrow$ *outcome*
 - **Temporal order**: past cannot depend on future
- These constraints are passed to the discovery algorithm as a “**knowledge**” **object** (Meek, 2006)
- **Pros**: faster, more accurate, more interpretable
- **Cons**: wrong priors inject wrong edges

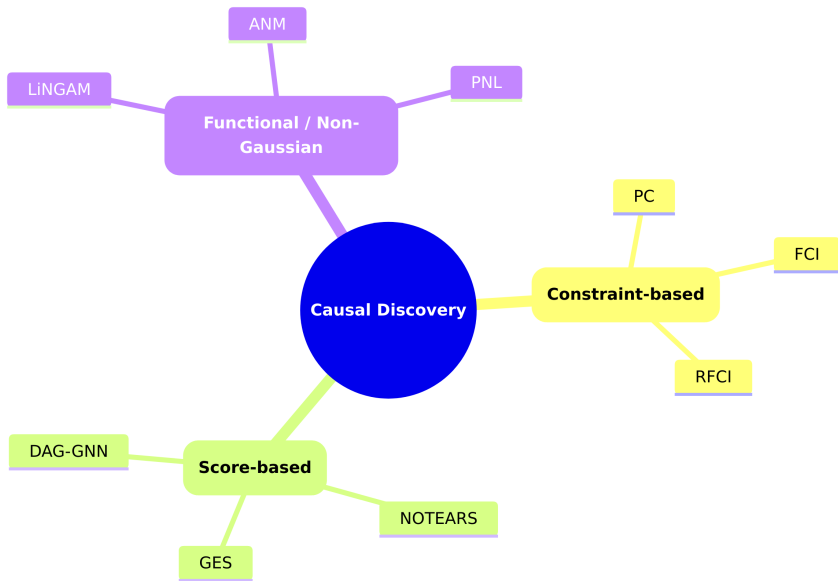
Combining Automated Discovery with Expert Judgment

- **Typical workflow:**
 1. Expert provides *variable list* and *temporal ordering*
 2. Expert marks *forbidden* and *required* edges
 3. Algorithm proposes a CPDAG over remaining edges
 4. Expert reviews output; refutes edges via subject-matter reasoning
 5. Active learning suggests *next intervention* to disambiguate edges
- **Example:** *Active causal structure learning* (Squires et al., 2018)
 - Pick the intervention that *most reduces* the number of DAGs consistent with the data
 - Analogous to active learning in classification



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- ***Discovery Algorithm Families***
 - Constraint-Based Methods: PC and FCI
 - Score-Based Methods: GES and NOTEARS
 - Non-Gaussian and Functional Methods
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Three Main Algorithm Families



Constraint-based uses conditional independence tests

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PC Algorithm: Idea

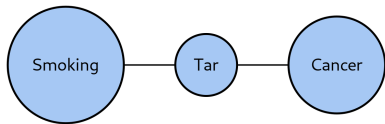
- Named after inventors **P**eter Spirtes and **C**larke Glymour (1993)
- **Key idea**: use conditional independence (CI) tests to recover the skeleton of the DAG, then orient edges
- **Input**: dataset $\mathcal{D}$, CI test (e.g., partial correlation, G^2 -test)
- **Output**: a CPDAG representing the Markov equivalence class
- **Assumes**:
 - Causal sufficiency (no hidden confounders)
 - Faithfulness
 - Correct CI tests

PC Algorithm: Steps

- **Steps:**
 1. Start from a fully connected undirected graph over $X_1, \dots, X_n$
 2. For each pair (X_i, X_j) , test $X_i \perp X_j \mid S$ for subsets S of increasing size
 - If independence found, remove the edge and record the *separating set* S_{ij}
 3. Identify **v-structures**: for any unshielded triple $X_i - X_k - X_j$, if $X_k \notin S_{ij}$ then orient as $X_i \rightarrow X_k \leftarrow X_j$
 4. Apply **orientation rules** (Meek) to propagate directions without creating new v-structures or cycles
- **Complexity:** $O(n^q)$ in the number of CI tests, where q is the maximum degree of the true DAG

PC Algorithm: Example

- **Variables:** Smoking (S), Tar (T), Cancer (C)
- **CI tests:**
 - $S \not\perp C$
 - $S \not\perp T$
 - $T \not\perp C$
 - $S \perp C \mid T \Rightarrow$ separating set $\{T\}$
- **Conclusion:**
 - Remove edge $S - C$
 - No v-structure (T is in separating set)
 - Skeleton: $S - T - C$
- Directions of edges $S - T - C$ remain **undirected** without extra assumptions



FCI: Dealing with Latent Confounders

- **PC fails** under hidden confounders (causal sufficiency violated)
- **FCI (Fast Causal Inference)** (Spirtes et al., 1995; Zhang, 1999)
 - Extends PC to handle *unobserved* common causes and selection bias
 - Output: a **PAG** (Partial Ancestral Graph), with edge marks:
 - $\rightarrow$: causes
 - $\leftrightarrow$: confounded (unobserved common cause)
 - $\circ\rightarrow$: ambiguous
 - $\circ\text{---}\circ$: undetermined
- **Pros**: sound under latent confounders
- **Cons**: slower; output less informative (many $\circ$ marks)
- **Example**: in econometrics with unobserved “ability” confounding education and wages, FCI correctly marks Education $\leftrightarrow$ Wage instead of asserting causation

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Score-Based Approach: Idea

- Treat discovery as a **search problem**:
 - Space: all DAGs G over n nodes
 - Objective: score $S(G, \mathcal{D})$ measuring how well G explains the data
- **Common scores**:
 - **BIC** (Bayesian Information Criterion):

$$S(G) = \log \Pr(\mathcal{D} \mid G, \hat{\theta}) - \frac{k}{2} \log N$$

where k = number of parameters, N = sample size

- **BDeu** (Bayesian Dirichlet equivalent uniform)
- **Pros**: sound theory, decomposes over nodes
- **Cons**: DAG space is super-exponential $\rightarrow$ heuristic search

GES: Greedy Equivalence Search

- **GES** (Chickering, 2002)
- **Input**: dataset $\mathcal{D}$, decomposable score $S(\cdot)$
- **Output**: a CPDAG (equivalence class)
- **Steps**:
 1. **Forward phase**: start from empty graph; add the edge that most *increases* the score, repeat until no improvement
 2. **Backward phase**: remove the edge that most *increases* the score, until no improvement
- Proven to recover the true equivalence class in the **large-sample limit**
- **Complexity**: polynomial per step, but many steps; practical up to $n \approx 100$ variables

NOTEARS: Continuous Optimization

- **Key idea** (Zheng et al., 2018): turn discrete DAG search into a **continuous optimization** problem
- Represent the graph by a weighted adjacency matrix $W \in \mathbb{R}^{n \times n}$
- DAG-ness constraint encoded as a smooth function:

$$h(W) = \text{tr}(e^{W \odot W}) - n = 0$$

where $h(W) = 0$ iff W corresponds to a DAG

- **Problem:**

$$\min_W \|X - XW\|_F^2 + \lambda \|W\|_1 \quad \text{s.t.} \quad h(W) = 0$$

- **Pros:** solvable with gradient descent; scales to 100s of variables
- **Cons:** sensitive to data scaling (Reisach et al., 2021); local minima

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LiNGAM: Linear Non-Gaussian Acyclic Model

- **Assumption:** data are generated by

$$X_i = \sum_j b_{ij} X_j + e_i$$

where

- b_{ij} encodes a DAG (acyclic B)
- Noises e_i are *mutually independent* and *non-Gaussian*
- **Key insight:** *non-Gaussian* noise breaks the symmetry
 - Gaussian case: $X \rightarrow Y$ and $Y \rightarrow X$ are indistinguishable
 - Non-Gaussian: asymmetry $\Rightarrow$ direction is **identifiable**
- Algorithm uses **Independent Component Analysis (ICA)** to recover B
- **Example:** in a stock-returns panel, non-Gaussian “fat tails” allow LiNGAM to orient edges that Gaussian methods cannot

Discovery Algorithm Families: Summary

Family	Example	Uses	Output	Key limit
Constraint-based	PC, FCI	CI tests	CPDAG/PAG	Sample size for CI tests
Score-based	GES, NOTEARS	Likelihood + regularization	CPDAG / W	Search / optimization hardness
Functional	LiNGAM, ANM	Non-Gaussianity, nonlinearity	Full DAG	Strong functional assumptions
Interventional	IGSP, JCI	Data from interventions	Full DAG	Cost of experiments

- **Recommendation:**
 - Start with **PC** or **GES** for exploratory analysis
 - Use **LiNGAM** when non-Gaussianity is plausible
 - Use **FCI** when latent confounders are likely
 - Use **interventional** methods when experiments are affordable

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 - Multiple Models Produce Identical Predictions
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 - When Discovery Should Change Your DAG

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Why Standard ML Fails at Causal Discovery

- Standard ML optimizes **predictive accuracy**, not causal structure
 - Correlation suffices for prediction
 - Direction of arrows is irrelevant to loss
- **Symptoms of the mismatch:**
 - Models exploit *shortcuts* (spurious features)
 - Fail under distribution shift
 - Cannot answer “*what if we intervene?*” queries
- **Example:** a classifier trained to detect cows may rely on grass backgrounds rather than cow features
 - High test accuracy, zero causal understanding
 - Fails on cows at the beach
- **Scholkopf (2019):** “*Causality for Machine Learning*” argues that to build robust AI, we must reason about *causal* mechanisms, not just correlations

Standard ML vs. Causal Discovery

Standard ML

- Goal: minimize prediction error
- Assumes i.i.d. data
- Optimizes $\Pr(Y | X)$
- Blind to arrow direction
- Fails under shift
- **Pros**
 - Simple, scalable
 - Mature tooling
- **Cons**
 - No interventions
 - No counterfactuals

Causal Discovery

- Goal: recover G
- Assumes structural model
- Targets $\Pr(Y | do(X))$
- Edge directions matter
- Stable under shift
- **Pros**
 - Supports intervention
 - Interpretable
- **Cons**
 - Assumption-heavy
 - Harder to validate

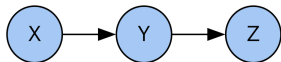
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The Identifiability Ceiling

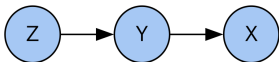
- **Observation:** many DAGs induce the **same joint distribution**
 - They are *predictively equivalent*
 - They are *causally* different
- Three classic patterns:
 1. **Chain:** $X \rightarrow Y \rightarrow Z$
 2. **Reverse chain:** $Z \rightarrow Y \rightarrow X$
 3. **Common cause:** $X \leftarrow Y \rightarrow Z$
- All imply $X \perp Z \mid Y$ and are **indistinguishable** from observational data
- **Implication:** no matter how much data you collect, you cannot identify which of these holds without either:
 - An intervention
 - A functional assumption (e.g., non-Gaussianity)

Markov Equivalence: Example

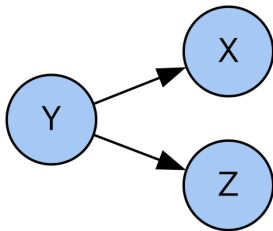
Chain 1



Chain 2



Common cause



- All three imply $X \perp Z \mid Y$
- Observational distribution identical
- **Causal implications are very different:**
 - Intervening on X in Chain 1 affects Z
 - Intervening on X in Chain 2 does *not* affect Z
 - Intervening on X in Common Cause does *not* affect Z

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Validating a Discovered DAG

- A recovered DAG is a *hypothesis* until validated
- **Validation strategies:**
 1. **Expert review:** do edges match known mechanisms?
 2. **Refutation tests:** does data contradict implied conditional independencies?
 3. **Sensitivity analysis:** do results change under perturbations of the assumptions?
 4. **Interventional check:** do targeted experiments confirm predicted effects?
- **Tools:** `causalnex`, `dowhy`, `CausalDiscoveryToolbox`, `gCastle`

Refutation Testing

- Given a discovered DAG G , check each implied independence against data
- **Procedure:**
 1. Enumerate CI statements implied by G (via d-separation)
 2. Test each on held-out data using an appropriate CI test
 3. Count violations; low count $\Rightarrow$ DAG is *consistent*
- **Placebo tests** (Pearl): introduce a known-null edge; discovery should assign it low weight
- **Example** (DoWhy library):
 - Replace treatment with a random variable; effect should vanish
 - Add an unobserved confounder; estimate sensitivity of conclusions
- **Q:** what if the DAG passes all refutation tests?
 - **Answer:** it is *consistent*, not *proven*. Causality is never established by data alone; always requires domain-informed interpretation

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When Discovery Should Change Your DAG

- **Update your DAG when:**
 - Multiple independent algorithms converge on the same edge
 - The new edge is *mechanistically plausible*
 - Refutation tests fail your current DAG
 - Interventional experiments confirm the discovered structure
- **Do *not* update your DAG when:**
 - Edge contradicts well-established mechanism (e.g., time order, physics)
 - Result hinges on weak faithfulness near independence boundary
 - Sample size is too small for reliable CI tests
 - Hidden confounders could explain the edge
- **Rule of thumb:** discovery shifts the *prior probability* of an edge; it does not provide a *posterior certainty*

Causal Discovery: Key Takeaways

- **Causal discovery** infers DAG structure from data, but only under strong assumptions (Markov, faithfulness, sufficiency)
- **Identifiability is limited**
 - Observational data recover only the Markov equivalence class
 - Interventions or functional assumptions are needed for full identification
- **Three algorithm families:**
 - Constraint-based (PC, FCI)
 - Score-based (GES, NOTEARS)
 - Functional / non-Gaussian (LiNGAM)
- **Discovery $\neq$ truth**
 - Use it to *generate* hypotheses
 - Combine with domain knowledge and interventions
 - Validate via refutation and expert review
- **Bottom line:** automated discovery is a powerful exploratory tool, not a replacement for scientific reasoning